Answer key. Answers are in magenta.

NRES 746, Week 8, Lecture 2: Curvature by Hand (optional take-home)

Optional practice; you don't need to turn it in. Try Parts 1 and 2 on paper before you open a laptop. Part 3 uses R.

Putting it in context

In Week 7 we saw that the curvature of the negative log-likelihood (NLL) at its minimum measures how much the data tell us about a parameter. For a parameter θ with MLE $\hat{\theta}$:

$$\mathrm{SE}(\hat{\theta}) \approx \frac{1}{\sqrt{\mathrm{NLL}''(\hat{\theta})}} \qquad \text{approximate 95\% interval: } \hat{\theta} \pm 1.96\,\mathrm{SE}$$

`optim(..., hessian = TRUE)` estimates $\mathrm{NLL}''(\hat{\theta})$ numerically. Here you find it with calculus, for two models you already know. (Part 1 starts where the Newton's method take-home did, then goes on to the standard error.)

Part 1: Pika haypiles (Poisson)

The Week 7 pika data: active haypiles in six talus plots, 2, 5, 3, 0, 4, 4, modeled as Poisson with mean λ. Dropping terms that don't involve λ,

$$\mathrm{NLL}(\lambda) = n\lambda - \left(\sum_{i=1}^{n} y_i\right) \log \lambda$$

1a. Find $\mathrm{NLL}'(\lambda)$ and $\mathrm{NLL}''(\lambda)$.

$\mathrm{NLL}'(\lambda) = n - \dfrac{\sum y_i}{\lambda}, \qquad \mathrm{NLL}''(\lambda) = \dfrac{\sum y_i}{\lambda^2}$

1b. Evaluate $\mathrm{NLL}''(\hat{\lambda})$ at $\hat{\lambda} = 3$ for the six plots, and again for 24 plots with the same mean (each count appearing four times). Give the SE for each.

6 plots: $\mathrm{NLL}''(\hat{\lambda}) = 18/9 = 2$, $\mathrm{SE} = 1/\sqrt{2} \approx 0.71$ 24 plots: $\mathrm{NLL}''(\hat{\lambda}) = 72/9 = 8$, $\mathrm{SE} = 1/\sqrt{8} \approx 0.35$

1c. Substitute $\hat{\lambda} = \bar{y} = \sum y_i / n$ into your NLL'' and simplify to get a general formula for $\mathrm{SE}(\hat{\lambda})$ in terms of $\hat{\lambda}$ and n. What happens to the SE when n quadruples?

$\mathrm{NLL}''(\hat{\lambda}) = \dfrac{n\hat{\lambda}}{\hat{\lambda}^2} = \dfrac{n}{\hat{\lambda}}$, so $\mathrm{SE}(\hat{\lambda}) = \sqrt{\hat{\lambda}/n}$. Quadrupling n halves the SE.

Part 2: Frog detection (binomial)

In Lab 5, Act 1, you pooled three wetlands and heard the frog on $K = 12$ of $N = 30$ visits, so $\hat{p} = 0.4$. Dropping the binomial coefficients,

$$\mathrm{NLL}(p) = -K \log p - (N - K) \log(1 - p)$$

2a. Find $\mathrm{NLL}''(p)$. (Watch the chain rule on $\log(1 - p)$.)

$\mathrm{NLL}'(p) = -\dfrac{K}{p} + \dfrac{N - K}{1 - p}, \qquad \mathrm{NLL}''(p) = \dfrac{K}{p^2} + \dfrac{N - K}{(1 - p)^2}$

2b. Evaluate $\mathrm{NLL}''(\hat{p})$ and the SE for your survey ($K = 12$, $N = 30$) and for your colleague's survey ($K = 120$, $N = 300$). Check: one of your curvatures should be 125.

Curvatures: $12/0.16 + 18/0.36 = 125$ (3 wetlands) and 1250 (30 wetlands).

3 wetlands: SE $= 1/\sqrt{125} \approx 0.089$ 30 wetlands: SE $= 1/\sqrt{1250} \approx 0.0283$

2c. Substitute $\hat{p} = K/N$ and simplify to get a general formula for SE($\hat{p}$). Have you seen this formula before?

$\text{NLL}''(\hat{p}) = \dfrac{N}{\hat{p}(1-\hat{p})}$, so $\text{SE}(\hat{p}) = \sqrt{\dfrac{\hat{p}(1-\hat{p})}{N}}$.

Yes: the standard error of a proportion from intro stats.

2d. The Week 7 Monday WOD (*Reading the Interval*) had 2 chytrid-positive frogs out of 20. Use your formula from 2c to build the approximate 95% interval, $\hat{p} \pm 1.96\,\text{SE}$. What's wrong with it? Compare it with the range you read off the NLL curve in that WOD.

Interval: from -0.031 to 0.231

The lower end is a negative prevalence, which is impossible. The NLL-curve range from that WOD (about 0.02 to 0.28) stays positive and is lopsided toward higher values.

Part 3: Check it in R

Using AI: the coding question below is labeled with a recommendation. Writing this one yourself is good practice for the labs.

3a. *(Recommended: no AI.)* Write an NLL function for the frog data in 2b ($K = 12$, $N = 30$) using `dbinom()`, and minimize it with `optim(..., method = "Brent", lower = 0.001, upper = 0.999, hessian = TRUE)`. Does the Hessian match your answer to 2b?

`optim()` Hessian: about 125 (yes, it matches 2b)

3b. Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest?

Open-ended.